FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – II

JEE (Main)-2024

TEST DATE: 20-01-2024

ANSWERS, HINTS & SOLUTIONS

Physics

PART – A

SECTION – A

1. B

Sol. $(V_1 - V_2)^2 - 0 = 2a - s$

$S = \dfrac{(V_1 - V_2)^2}{2a}$

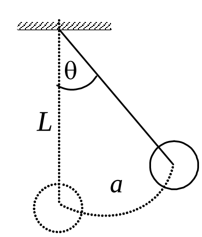
$S < d$

$\dfrac{(V_1 - V_2)^2}{2d} < d$

2. B

Sol. For $T_{max.}$, θ = 0
at lowest point

$T = mg + \dfrac{mv^2_{\max i}}{L} = mg + \dfrac{ma^2\omega^2}{L} = mg\left[1 + \left(\dfrac{a}{L}\right)^2\right]$



3. D

Sol. $E = C_1C_2 \cos\omega_0 t + \dfrac{C_1C_3}{2}\cos(\omega_0 + \omega)t + \dfrac{C_1C_3}{2}\cos(\omega_0 - \omega)t$

Of the three components, the highest frequency component will liberate the electrons with maximum kinetic energy

$(K.E)_{\max} + \phi = \dfrac{h}{2\pi}(\omega_0 + \omega) \Rightarrow \phi = 2.39$ eV.

4. B

Sol. Let x $\rightarrow$ length outside liquid

Initially, $\sigma_w A\ell = \rho A(\ell - x)$

$$\frac{\sigma}{\ell} = \frac{\ell \cdot x}{\ell}$$

$$\frac{1}{2} = \frac{\ell - x}{\ell}$$

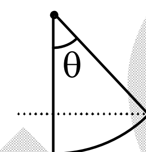
$$x = \frac{\ell}{2}$$

After 1 hour

$$\frac{(\ell-2)-(x+y)}{\ell-2} = \frac{1}{2}$$

$$y = -1$$

5. B



Sol. $$\sin\theta = \frac{\frac{mv}{\sqrt{2}\,qB}}{\frac{mv}{qB}} = \frac{1}{\sqrt{2}}$$

$\Rightarrow\ \theta = 45°$

$$t = \frac{T}{8} = \frac{\pi m}{4qB}$$

6. A

Sol. Stopping distance $\propto v^2$

$\therefore$ v has increased by factor of 2

7. C

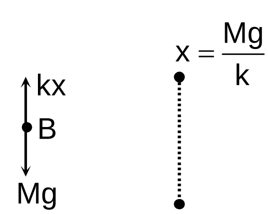
Sol. Increment in the length AB = $\frac{MgL}{AY} = \frac{10\times10\times0.1}{10^{-4}\times2.5\times10^{10}} = 4\times10^{-6}\,m$

$\therefore$ Displacement of point B = $4\times10^{-6}\,m$

8. A

Sol. For A $\frac{1}{2}kx^2 = mgx$

$$m = \frac{M}{2}$$



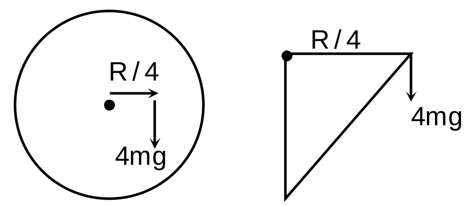
9. B

Sol. Locate C.M.

$$x_{cm} = \frac{O(3m)+mR}{4m}$$

$x_{cm} = \frac{R}{4}$ $\tau_{net} = I_{sys}\,\alpha$

Along x-axis $4mg\left(\frac{R}{4}\right) - fR = I\,\alpha$



fr = 4ma $\quad\quad \alpha = \dfrac{g}{8R}$

fr = 4 m & R

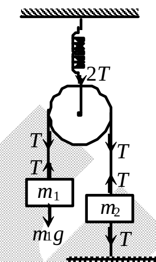
10. A

Sol. $T = m_1 g$

$kx = 2T = 2m_1 g$

$x = \dfrac{2m_1 g}{k}$

Energy stored = $\dfrac{1}{2}kx^2 = \dfrac{1}{2}k\dfrac{4m_1^{\,2}g^2}{k^2} = \dfrac{2m_1^{\,2}g^2}{k}$



11. B

Sol. $JL = I\omega$; $\quad \omega = \dfrac{JL}{I} = \dfrac{10\times 2}{\frac{1\times 2^2}{3}} = \dfrac{60}{4} = 15$ rad/s

12. B

Sol. % error in density = $\left[2\left(\dfrac{0.005}{0.5}\right) + \dfrac{0.003}{0.3} + \dfrac{0.06}{6}\right]\times 100$ = 4

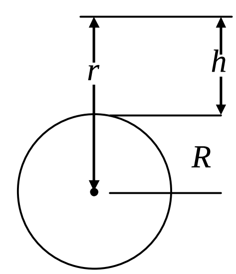
13. C

Sol. By conservation of energy,

$\dfrac{1}{2}mv^2 = \dfrac{mgh}{\left(1+\dfrac{h}{R}\right)}$

here $\quad v = kv_e = k\sqrt{2gR}$

$\therefore \quad \dfrac{1}{2}m(k\sqrt{2gR})^2 = \dfrac{mg\,h}{1+h/R}, \quad h = \dfrac{k^2R}{1-k^2}$



14. B

Sol. $V = Blv$, $\quad Q = CV = BlvC$

15. A

Sol. $\eta = 1 - \dfrac{T_2}{T_1}$

$\dfrac{1}{5} = 1 - \dfrac{T_2}{T_1}$

$\dfrac{1}{4} = 1 - \dfrac{T'_2}{T_1}$

$\dfrac{T_2 - T'_2}{T_1} = \dfrac{1}{4} - \dfrac{1}{5} = \dfrac{1}{20}$

$$\Delta T_2 = \frac{1}{20} \times (330 + 273) = 30.15^{o}\,K$$

16. A

Sol. Let v is the final velocity of particle then

$$\frac{1}{2}mv_0^2 - \frac{3}{4}\left(\frac{1}{2}mv_0^2\right) = \frac{1}{2}mv^2$$

$$v = \frac{v_0}{2}$$

$\therefore \quad v = u + at$

$$\frac{v_0}{2} = v_0 - \mu g t_0 \Rightarrow \mu = \frac{v_0}{2gt_0}$$

17. C

Sol. $F = -\dfrac{dU}{dx} = -8\sin 2x$, $a = \dfrac{F}{m} = -8\sin 2x \qquad (m = 1kg)$

for small oscillations $\sin 2x \approx 2x$ i.e. $a = -16x$

since $a \propto -x$

oscillations are simple harmonic in nature

$\therefore \quad T = \dfrac{\pi}{2}s$

18. C

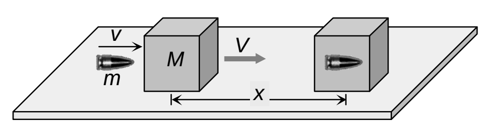
Sol. $C = \dfrac{M_1(0) + M_2(\ell)}{M_1 + M_2} = \dfrac{M_2\ell}{(M_1 + M_2)}$

$$C' = \frac{M_1\ell_1 + M_2(\ell + \ell_2)}{M_1 + M_2}$$

$$\therefore\ C' - C = \frac{M_1\ell_1 + M_2\ell_2}{M_1 + M_2}$$

19. C

Sol. Let speed of the bullet = v



Speed of the system after the collision = V

By conservation of momentum $mv = (m + M)V$

$$\Rightarrow V = \frac{mv}{M + m}$$

So the initial K.E. acquired by the system

$$= \frac{1}{2}(M + m)V^2 = \frac{1}{2}(m + M)\left(\frac{mv}{M + m}\right)^2 = \frac{1}{2}\frac{m^2v^2}{(m + M)}$$

This kinetic energy goes against friction work done by friction = $\mu R \times x = \mu(m + M)g \times x$

By the law of conservation of energy

$$\frac{1}{2}\frac{m^2v^2}{(m + M)} = \mu(m + M)g \times x \Rightarrow v^2 = 2\mu gx\left(\frac{m + M}{m}\right)^2$$

$$\therefore\ v = \sqrt{2\mu gx}\left(\frac{M + m}{m}\right)$$

20. B

Sol. $g = \frac{GM}{R^2} \Rightarrow \frac{dg}{g} = -2\frac{dR}{R}$

$\frac{dR}{R} = -1\% \Rightarrow \frac{dg}{g} = 2\%$

SECTION – B

21. 8

Sol. The frequency of transverse waves in a stretched string is given by

$n = \frac{p}{2\ell}\sqrt{\frac{T}{m}}$,

$\frac{p}{2\ell_A} = \frac{q}{2\ell_B}\sqrt{\frac{T}{m_B}}$

$\therefore \quad \frac{p}{q} = \frac{\ell_A}{\ell_B}\sqrt{\frac{m_A}{m_B}} = \frac{\ell_A}{\ell_B}\sqrt{\frac{\rho_A}{\rho_B}}$

$= \frac{0.3}{0.75}\sqrt{\frac{6.3}{2.8}} = \frac{3}{5}$

So, $p = 3, q = 5$

No. of antinodes = p + q = 3 + 5 = 8

22. 1

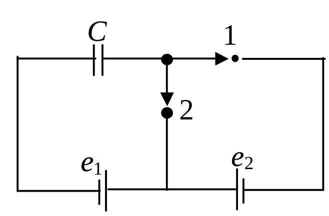
Sol. Volume flow rate Q = $\int(2\pi rv)dr$

$= \int_0^R 2\pi v_0 r\left[\frac{R^2 - r^2}{R^2}\right]dr$

$= 2\pi v_0\left[\frac{R^2}{2} - \frac{R^2}{4}\right] = \frac{\pi v_0 R^2}{2}$

23. 100

Sol. In position 1



Stored energy = $\frac{1}{2}C(e_1 - e_2)^2$

In position 2

Stored energy = $\frac{1}{2}Ce_1^2$

Extra energy drawn from the battery = $e_1\,\Delta q$ where Δq is the additional charge drawn from the battery.

Now $\Delta q = e_1C - (e_1 - e_2)C = e_2C$

$\therefore$ Extra energy drawn from the battery = $e_1(e_2C) = e_1e_2C$

Heat produced = loss in stored battery + extra energy drawn from the battery

$$= \frac{1}{2}(e_1 - e_2)^2C - \frac{1}{2}e_1^2C + e_1e_2C = \frac{1}{2}e_1^2C - e_1e_2C + \frac{1}{2}e_2^2C - \frac{1}{2}e_1^2C + e_1e_2C = \frac{1}{2}e_2^2C$$

$$= \frac{1}{2}\times 2\mu F\times(10)^2 = 100\ \mu J$$

24. 200

Sol. At any time length of each wire $= l - 2vt$

Induced emf $= 4\ Bv(l - 2vt)$

Induced current $= \dfrac{4Bv(l-2vt)}{4\lambda(l-2vt)} = \dfrac{Bv}{\lambda}$, $F = B\left(\dfrac{Bv}{\lambda}\right)(l-2vt) = \dfrac{B^2v}{\lambda}(l-2vt)$

$= \dfrac{(2)^2 \times 5}{0.5}(15 - 2\times 5\times 1)$ = 200 N

25. 942

Sol. $Z^2 = R^2 + (\omega L - \dfrac{1}{\omega C})^2$ … (i)

$\cos\phi = \dfrac{R}{Z} = 0.8$

$\therefore$ $R = 0.8Z$ … (ii)

Putting this value in equation (i)

$0.64\ Z^2 + (\omega L - \dfrac{1}{\omega C})^2 = Z^2$

$\therefore$ $\omega L - \dfrac{1}{\omega C} = 0.6Z$ … (iii)

Also given $V_C = 2/5\ V_L$

$I\ X_C = 2/5\ X_L I$

$\therefore$ $X_C = 0.4X_L$ … (iv)

$\therefore$ Putting this in equation (iii)

$\omega L - 0.4\omega L = Z$

$\therefore$ $Z = 0.6\ \omega L = 0.6 \times 5 \times 314$

$Z = 942\ \Omega$

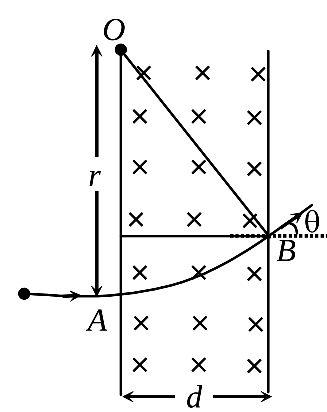
26. 30

Sol. When an α-particle is accelerated by 10^4 volt, its kinetic energy will be $K = (2e)(10^4\text{ V}) = 2\times 10^4 e$ V

Now the path of a charged particle when it enters a magnetic field at right angels is a circle with radius

$$r = \frac{mv}{qB} = \frac{\sqrt{2mK}}{qB}$$

So here, $r = \dfrac{\left(2\times 6.4\times 10^{-27}\times 2\times 10^4\times 1.6\times 10^{-19}\right)^{\frac{1}{2}}}{2\times 1.6\times 10^{-19}\times 0.1}$ = 0.2 m



Now as in case of a circle angle between tangents at two points will be equal to the angle between normals at these points. As in a circle tangent is normal to radius at every point, the change in direction of the particle as it passes the field.

$$\theta = \sin^{-1}\left(\frac{d}{r}\right) = \sin^{-1}\left(\frac{0.1}{0.2}\right) = 30^0$$

27. 5

Sol. Let V be the initial volume of the balloon, and M be its mass.

Weight of water displaced = weight of balloon

i.e. $\frac{2}{3}V\rho g = Mg$ or $M = \frac{2}{3}V\rho$

Now, let the depth of immersion be h m, so that the new pressure = (10 + h) m of water.

Applying Boyle's law $P_1V_1 = P_2V_2$

i.e., $10V = (10+h)V'$

$M = V'\rho$

$\therefore$ $h = 5$ m

28. 1

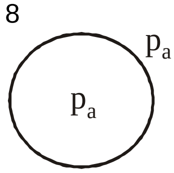
Sol. Taking moments at out pt. A

$(6-x)^2 A p_w g/2 = 18 A p_r g$

Solving we get $x = 1$ m.

29. 8

Sol.



Inside pressure must be $\frac{4T}{r}$ greater than outside pressure in bubble. This excess pressure is provided by charge on bubble.

$\frac{4T}{r} = \frac{\sigma^2}{2\varepsilon_0}$; $\frac{4T}{r} = \frac{Q^2}{16\pi^2 r^2 \times 2\varepsilon_0}$ $\left[\sigma = \frac{Q}{4\pi r^2}\right]$

$Q = 8\pi r\sqrt{2rT\varepsilon_0}$

30. 2

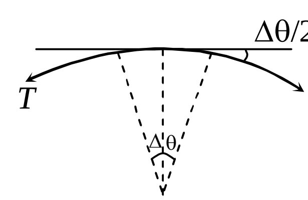
Sol.

$2T\sin\frac{\Delta\theta}{2} = dm\omega^2 r$

$2T\left(\frac{\Delta\theta}{2}\right) = \rho \times A \times r\Delta\theta \times \omega^2 \times r$

$\sigma = \frac{T}{A} = \rho r^2\omega^2$

$\therefore$ $\omega = \frac{1}{r}\sqrt{\frac{\sigma}{\rho}}$ = 2 rad/s



Chemistry

PART – B

SECTION – A

31. B

Sol. $E = \frac{hc}{\lambda} = 2.9\times10^{-19}\,J$

% efficiency = $\frac{\text{energy stored}}{\text{Total energy of lognata}}\times100$

32. A

Sol. $\Delta G^o = -nFE^{\circ}_{cell}$

$E^o = \frac{nE^{\circ}_1 + n_2E^{\circ}_2}{n_3}$

33. B

Sol. VSEPR theory repulsion between CH_3, CH_3 is greater than H. H atoms

34. B

Sol. $\Delta T_f = K_f m \quad K_f = \frac{M_{camphor}RT_f^{02}}{\Delta_{fus}H_{1,m}}$

35. A

Sol. $\Delta_r G^o = -nRT\ln K^{\circ}_p$

36. C

Sol. $H_2\ddot{O}:$ base which accept H^+

$H_2\ddot{O}:$ Lewis base

$\underset{\text{Reductant}}{\underset{\downarrow}{H_2O^{-2}}} \rightarrow O^{\circ}_2$

37. C

Sol. 1 mole of $NaCO_3$ gives 2 mole of Na^+

38. A

Sol. $Ba(IO_3) \rightleftharpoons Ba^{2+} + 2IO_3^-$

$K_{sp} = [Ba^{2+}][IO_3^-]^2$

39. C

Sol. Ionic compound –soluble in aqueous
Covalent compound –soluble in organic solvent

40. B

Sol. Indicator theory, phenolphthalein use for strong base titration

41. C
Sol. PPh_3 increase e^- density of metal this will increase backdonation

42. A
Sol. g = gerads, u = ungerads

43. A
Sol. Geometrical isomers arise because of change in bond angle

44. D
Sol. –205 – 152 = –357

Enthalpy of hydrogenation = $= \frac{-357}{3} = -119\,kJ/mol$

45. C
Sol. 4(4n + 2) πe^- follow – aromatic

46. B
Sol. $t_{1/2} = \frac{0.693}{\lambda_1 + \lambda_2}$

% yield of $Th^{227} = \frac{\lambda_1}{\lambda_1 + \lambda_2} \times 100$

% yield of $Fr^{223} = \frac{\lambda_2}{\lambda_1 + \lambda_2} \times 100$

47. C
Sol. Group properties

48. A
Sol. f block radii $La^{2+} > Eu^{3+} > Gd^{3+} > Lu^{3+}$

49. D
Sol. Factual

50. B
Sol. 0.1 M complex = 0.1 mole in 1 L; 28.7g = 0.2 mole AgCl

SECTION – B

51. 2
Sol. Graph of weak acid v/s strong base

52. 5
Sol. Rate of reaction = $K[CO]^x[O_2]^y$

$4 \times 10^{-5} = K(0.02)^x (0.02)^y$ …(i)

$1.6 \times 10^{-4} = K (0.04)^x (0.02)^y$ …(ii)

$8 \times 10^{-5} = K(0.02)^x (0.04)^y$ …(iii)

$\frac{(i)}{(ii)}\ \frac{1}{4} = \left(\frac{1}{2}\right)^x$

$\frac{(i)}{(ii)}\ \frac{1}{2} = \left(\frac{1}{2}\right)^y$

$y = 1$

From eq. (1)

$4\times10^{-5} = K(0.02)^2(0.02)^1$

$K = \dfrac{4\times10^{-5}}{4\times10^{-4}\times2\times10^{-2}} = \dfrac{10}{2} = 5$

53. 4

Sol. $H_2CO_3 + NaOH \rightarrow NaHCO_3 + H_2O$

$pH_1 = \frac{1}{2}\left(pK_{a_1} + pK_{a_2}\right) = \frac{1}{2}(4.6+8) = 6.3$

$N_1V_1 = N_2V_2$

$N_1\times10 = 0.1\times20$

$N_1 = 0.2$

$NaHCO_3 + NaOH \rightarrow Na_2CO_3 + H_2O$

$c = \dfrac{2\text{m mole}}{50\text{ mL}} = \dfrac{1}{25}M$

$pH_2 - pH_1 = 10.3 - 6.3 = 4$

54. 8

Sol. Stability of carbocation depend on hyper conjugation

55. 4

Sol. factual

56. 8

Sol. $\overset{\infty}{\Lambda}CH_3COOH = \overset{\infty}{\Lambda}HCl + \overset{\infty}{\Lambda}CH_3COOK - \overset{\infty}{\Lambda}KCl$

= 0.038 + 0.009 – 0.013

= 0.034

$= 100\% = \left(\dfrac{\Lambda}{\Lambda^\infty}\right)100$

$\dfrac{2.72\times10^{-3}}{0.034}\times100 = 8$

57. 7

Sol. –50 = $(\Delta_{Ion}H)_{NH_4OH} - 57$

$(\Delta_{Ion}H)_{NH_4OH} = 7kJ$

58. 6

Sol. Let mass of $K_2Cr_2O_7$ = a g

Mass of $KMnO_4$ = (10 – a)g

Eq. of $K_2Cr_2O_7$ + Eq. of $KMnO_4$ = Eq. of $Na_2S_2O_3$

$\left(\dfrac{a}{294}\times6\right) + \left(\dfrac{10-A}{158}\times5\right) = (2.2)(0.1)$

$a \approx 8.5\,g$

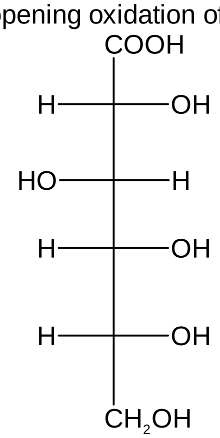
Mass % of $KMnO_4$ =Z = 15

59. 3

Sol. H_2O weak filed ligand

60. 4

Sol. After ring opening oxidation of α-D-glucose leads to formation of (A).

A $\longrightarrow$ 

Mathematics PART – C

SECTION – A

61. D

Sol. $g'(x) = (f'((\tan x - 1)^2 + 3))2(\tan x - 1)\sec^2 x$

since $f''(x) > 0 \Rightarrow f'(x)$ is increasing

So, $f'((\tan x - 1)^2 + 3) > f'(3) = 0 \ \forall \ x \in \left(0, \ \frac{\pi}{4}\right) \cup \left(\frac{\pi}{4}, \ \frac{\pi}{2}\right)$

Also $(\tan x - 1) > 0 \in \left(\frac{\pi}{4}, \ \frac{\pi}{2}\right)$. So g (x) is increasing in $\left(\frac{\pi}{4}, \ \frac{\pi}{2}\right)$.

62. B

Sol. Solving 2 cos x = 3 tan x we get, $2 - 2\sin^2 x = 3\sin x$

$\Rightarrow \sin x = \frac{1}{2} \Rightarrow x = \frac{\pi}{6}$.

Required area = $\int_0^{\pi/6} (2\cos x - 3\tan x)dx = 2\sin x - 3\ln\sec x\Big|_0^{\pi/6} = 1 - 3\ln 2 + \frac{3}{2}\ln 3$.

63. A

Sol. P (exactly two of A, B, c occur)

$= P(B \cap C) + P(C \cap A) + P(A \cap B) - 3P(A \cap B \cap C)$

$= P(B).P(C) + P(C).P(A) + P(A).(B) - 3P(A).P(B).P(C)$

$= \frac{1}{2}.\frac{1}{4} + \frac{1}{4}.\frac{1}{3} + \frac{1}{3}.\frac{1}{2} - 3.\frac{1}{3}.\frac{1}{2}.\frac{1}{4} \qquad = \frac{1}{4}$

64. B

Sol. Centre of the required circle is the reflection of the point (0, 0) in the line y = mx + m.

Let C (h, k) be the centre of the reflected circle

$\Rightarrow \frac{k}{h} = -\frac{1}{m}$ ……(1)

and $\frac{k}{2} = m\frac{h}{2} + m$ ……(2)

$\Rightarrow k = m(-km) + 2m \Rightarrow k = \frac{2m}{1+m^2}$

$\therefore$ C (h, k) is $\left(-\frac{2m^2}{1+m^2}, \ \frac{2m}{1+m^2}\right)$.

65. B

Sol. $z_1 = 3 + 4i$, $z_2 = 4 + 3i$, $z_3 = 2\sqrt{6} + i$

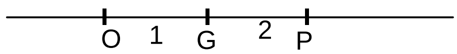
Clearly $|z_1| = |z_2| = |z_3| = 5$,

$\Rightarrow$ Points would lie on the circle centred at origin 'O'.

Now centroid of the triangle formed by these point

G = $\left(\frac{7 + 2\sqrt{6}}{3} + \frac{8i}{3}\right)$

$$OG = \sqrt{\left(\frac{7+2\sqrt{6}}{3}\right)^2 + \frac{64}{9}} = \frac{1}{3}\sqrt{137+28\sqrt{6}}$$



$\Rightarrow OP = 3\ OG = \sqrt{137+28\sqrt{6}}$.

66. B

Sol. We have $y = \sqrt{x\ z}$, so that the given determinant is equal to

$$\begin{vmatrix} \sqrt{x}\left(\sqrt{x}p+\sqrt{z}\right) & x & \sqrt{x\ z} \\ \sqrt{z}\left(\sqrt{x}\ p+\sqrt{z}\right) & \sqrt{xz} & z \\ 0 & \sqrt{x}\left(\sqrt{x}\ p+\sqrt{z}\right) & \sqrt{z}\left(\sqrt{x}\ p+\sqrt{z}\right) \end{vmatrix} = \sqrt{x}\ \sqrt{z}\left(\sqrt{x}\ p+\sqrt{z}\right)^2 \begin{vmatrix} 1 & \sqrt{x} & \sqrt{z} \\ 1 & \sqrt{x} & \sqrt{z} \\ 0 & \sqrt{x} & \sqrt{z} \end{vmatrix} = 0$$

67. A

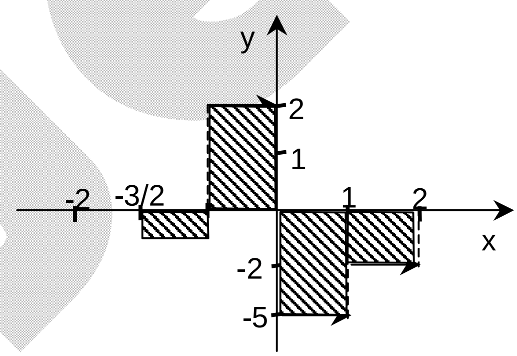
Sol. $3[x] - 5\dfrac{|x|}{x} = 3[x] - 5$, if $x > 0$

$= 3[x] + 5$, if $x < 0$

$$\Rightarrow \int_{-3/2}^{2} f(x)dx$$

$$= -1\left(-1+\frac{3}{2}\right) + 2(1) + 1(-5) + (-2)$$

$$= -\frac{1}{2} + 2 - 5 - 2 = -\frac{11}{2}.$$



68. D

Sol. $f(x) = (x^2 - 4)|(x^3 - 6x^2 + 11x - 6)| + \dfrac{x}{1+|x|}$

$\dfrac{x}{1+|x|}$ is always differentiable.

$(x - 2)(x + 2)|(x - 1)(x - 2)(x - 3)|$ is not differentiable at x = 1, 3.
So, f (x) is not differentiable at x = 1, 3.

69. A

Sol. Given equation can be written as
$x^2 - 3 = 3[\sin x]$
Case I: $x^2 - 3 = -3$ when $[\sin x] = -1$
$\Rightarrow x = 0$ but $\sin x \neq -1$
Case II: $x^2 - 3 = 0$ when $[\sin x] = 0$
$\Rightarrow x = \pm\sqrt{3} \Rightarrow x = \sqrt{3}$ gives $[\sin x] = 0$
Case III: $x^2 - 3 = 3$ when $[\sin x] = 1$
$\Rightarrow x = \pm\sqrt{6}$ but $[\sin x] \neq 1$.
Number of solution is one i.e. $x = \sqrt{3}$.

70. B

Sol. $(x-2)^2+(y+3)^2=\left(\frac{1}{2}\frac{3x-4y+7}{5}\right)^2$ is an ellipse, whose focus is (2, -3), directrix 3x – 4y + 7 = 0 and eccentricity is $\frac{1}{2}$.

Length of $\perp$ from focus to directrix is $\frac{3\times 2-4\times(-3)+7}{5}=5$

$\frac{a}{e}-ae=5 \quad \Rightarrow \quad 2a-\frac{a}{2}=5 \Rightarrow a=\frac{10}{3}$

So length of major axis is $\frac{20}{3}$

71. C

Sol. $f(x+y)-f(x-y)=(x+y)^2-(x-y)^2$

$\Rightarrow f(x+y)-(x+y)^2=f(x-y)-(x-y)^2$

$\Rightarrow f(x+y)=k+(x+y)^2$

$\Rightarrow f(x)=k+x^2$.

Since $f(0)=0 \Rightarrow f(x)=x^2$.

72. C

Sol. We have $\begin{bmatrix}\omega^2 & -\omega^2\\ \omega & \omega\end{bmatrix}^{-1}\begin{bmatrix}\omega & \omega^2\\ -\omega & \omega^2\end{bmatrix}^{-1}=\left(\begin{bmatrix}\omega & \omega^2\\ -\omega & \omega^2\end{bmatrix}\begin{bmatrix}\omega^2 & -\omega^2\\ \omega & \omega\end{bmatrix}\right)^{-1}=\begin{bmatrix}2 & 0\\ 0 & 2\end{bmatrix}^{-1}=\frac{1}{4}\begin{bmatrix}2 & 0\\ 0 & 2\end{bmatrix}$.

73. B

Sol. Given $|\vec{OA}+\vec{OB}|=|\vec{OA}+2\vec{OB}|$.

On squaring $(OA)^2+(OB)^2+2\vec{OA}.\vec{OB}=(OA)^2+4(OB)^2+4\vec{OA}.\vec{OB}$

$\Rightarrow 2\vec{OA}.\vec{OB}=-3(OB)^2<0$

$\Rightarrow 2|\vec{OA}|.|\vec{OB}|\cos\theta<0$

$\Rightarrow \cos\theta<0 \Rightarrow \theta>90°$

i.e. $\angle BOA>90°$.

74. A

Sol. Incentre $\left(\frac{ax_1+bx_2+cx_3}{a+b+c}, \frac{ay_1+by_2+cy_3}{a+b+c}\right)\equiv(0,\ 2-\sqrt{2})$

Here a = 2, b = $\sqrt{2}$, c = $\sqrt{2}$.

75. C

Sol. $f_n(x)=\tan\frac{x}{2}\left(\frac{1+\cos x}{\cos x}\right)(1+\sec 2x)(1+\sec 4x).......(1+\sec 2^n x)$

$=\tan x\left(\frac{1+\cos 2x}{\cos 2x}\right)(1+\sec 4x).......(1+\sec 2^n x)$

$=\tan 2x(1+\sec 2^2 x).......(1+\sec 2^n x)$

$=\tan 2^{n-1}x(1+\sec 2^n x)=\tan 2^n x$.

Now $\lim\limits_{x\to 0}\dfrac{f_n(x)}{2x} = \lim\limits_{x\to 0}\dfrac{\tan 2^n x}{2^n x}.2^{n-1} = 2^{n-1}$.

76. D

Sol. Given $\lim\limits_{x\to\infty} x\left[\tan^{-1}\left(\dfrac{2x+1}{2x+4}\right)-\dfrac{\pi}{4}\right] = \lim\limits_{x\to\infty} x\left[\tan^{-1}\dfrac{2x+1}{2x+4}-\tan^{-1}1\right]$

$$= \lim_{x\to\infty} x\tan^{-1}\left(\frac{\frac{2x+1}{2x+4}-1}{1+\frac{2x+1}{2x+4}}\right) = \lim_{x\to\infty} x\tan^{-1}\left(\frac{-3}{4x+5}\right) = \lim_{x\to\infty}\frac{\tan^{-1}\frac{-3}{4x+5}}{\frac{-3}{4x+5}}\times\frac{-3x}{4x+5} = -\frac{3}{4}$$

77. A

Sol. Let other end of diameter (h, k)

Hence centre is $\left(\dfrac{3+h}{2}, \dfrac{k+4}{2}\right)$. This circle touches x-axis means r = $\dfrac{k+4}{2}$

$= \sqrt{\left(\dfrac{3+h}{2}-3\right)^2+\left(\dfrac{k+4}{2}-4\right)^2}$ gives the equation of parabola.

78. C

Sol. Here $\log_x (x^2 - 2)$ to be defined

$x > 0$, $x \neq 1$ and $x^2 - 2 > 0 \Rightarrow x > \sqrt{2}$ or $x < -\sqrt{2}$.

$\Rightarrow x > \sqrt{2}$.

Now $\log_x (x^2 - 2) \geq 0$

$\Rightarrow x^2 - 2 \geq 1 \Rightarrow x^2 \geq 3$

$\Rightarrow x \geq \sqrt{3}$.

79. A

Sol. Any point on the parabola is $(x, x^2 + 7x + 2)$

Its distance from the line $y = 3x - 3$ is given by

P = $\left|\dfrac{3x-(x^2+7x+2)-3}{\sqrt{9+1}}\right| = \left|\dfrac{x^2+4x+5}{\sqrt{10}}\right| = \dfrac{x^2+4x+5}{\sqrt{10}}$ (as $x^2 + 4x + 5 > 0$ for all $x \in R$)

$\dfrac{dP}{dx} = 0 \Rightarrow x = -2$

The required point $\equiv (-2, -8)$.

80. B

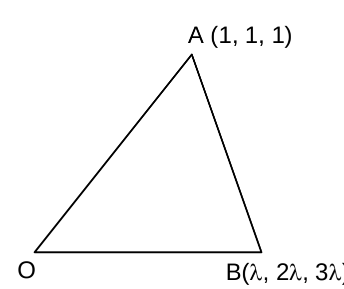
Sol. Let any point on second line be $(\lambda, 2\lambda, 3\lambda)$

$\cos\theta = \dfrac{6}{\sqrt{42}}$ $\sin\theta = \dfrac{6}{\sqrt{42}}$

$\Delta_{OAB} = \dfrac{1}{2}$ (OA). OB $\sin\theta = \dfrac{1}{2}\sqrt{3}\,.\,\lambda\sqrt{14}\times\dfrac{6}{\sqrt{42}} = \sqrt{6}$

$\Rightarrow \lambda = 2$

so B is (2, 4, 6)



SECTION – B

81. 36

Sol. $a^4 + \frac{1}{a^4} + 2 = 121 \Rightarrow a^2 + \frac{1}{a^2} = 11 \Rightarrow a - \frac{1}{a} = 3$

Now, $a^3 - \frac{1}{a^3} = \left(a - \frac{1}{a}\right)\left(a^2 + \frac{1}{a^2} + 1\right) = 3\,(11 + 1) = 36$

82. 1

Sol. Dividing the numerator and denominator by x^2, the given integral becomes

$$\int \frac{\left(1 - \left(\frac{1}{x^2}\right)\right)dx}{\left[\left(x + \left(\frac{1}{x}\right)\right)^2 + 1\right]\tan^{-1}\left(x + \left(\frac{1}{x}\right)\right)}$$

Let $x + \frac{1}{x} = \tan v = \int \frac{dv}{v} = \log|v| + c$

$= \log\left|\tan^{-1}\left(\frac{x^2+1}{x}\right)\right| + c$. Hence $k = 1$.

83. 5

Sol. For $(k-2)x^2 + 8x + k + 4 > 0$

$(k-2) > 0$ and $(8)^2 - 4(k-2)(k+4) < 0$

i.e. $k > 2$ ..... (1)

and $24 - k^2 - 2k < 0$

i.e. $k^2 + 2k - 24 > 0$

i.e. $(k+6)(k-4) > 0$

$\Rightarrow k < -6$ or $k > 4$ ..... (2)

From (1) and (2), we have $k > 4$.

$\therefore$ The least integral value of $k = 5$.

84. 1

Sol. Here $\sin^{-1}x - \cos^{-1}x = \pi/6$

Also $\sin^{-1}x + \cos^{-1}x = \pi/2$

$\Rightarrow \sin^{-1}x = \pi/3$ and $\cos^{-1}x = \pi/6$

$\Rightarrow x = \frac{\sqrt{3}}{2}$

85. 0

Sol. $\sin\theta + \sin\phi = \sqrt{3}(\cos\phi - \cos\theta)$

$\Rightarrow \sin\frac{\theta+\phi}{2}\cos\frac{\theta-\phi}{2} = \sqrt{3}\,\sin\frac{\theta+\phi}{2}\sin\frac{\theta-\phi}{2}$

$\Rightarrow$ either $\theta + \phi = 2n\pi$ or $\theta = 60° + \phi + 2n\pi$

Now $\sin 3\theta + \sin 3\phi = \sin 3\theta - \sin 3\theta = 0$

Or $\sin 3\theta + \sin 3\phi = \sin 3\theta - \sin(180° - 3\theta) = 0$

86. 25

Sol. $\vec{a}\cdot\vec{b} = 0 \Rightarrow x_1 + x_2 + x_3 = 0$

Thus we have to obtain the number of integral solution of this equation.

Coefficient of x° | $(x^{-3} + x^{-2} + x^{-1} + x^0 + x + x^2)^3$

$$= x^\circ \left| \left(\frac{1 + x + x^2 + x^3 + x^4 + x^5}{x^3} \right)^3 \right.$$

$= x^9 \,|(1 - x^6)^3 (1 - x)^{-3} = {}^{11}C_9 - 3.{}^5C_3 = 25$.

87. 4

Sol. $\tan^4 x - 2\tan^3 x - \tan^2 x + 2\tan x + 1$
$\tan^4 x + \tan^2 x - 2\tan^3 x + 2\tan x - 2\tan^2 x + 1$
$= (\tan^2 x - \tan x)^2 + 2(\tan x - \tan^2 x) + 1 = 4$.

88. 6

Sol. If a, b, c, d are in increasing order
$a = b^2 - c^2 + d^2 \Rightarrow b^2 + (d + c) = a \Rightarrow b^2 + d + 2 = 0$
which is impossible as b, d > 0.
If a, b, c, d are in decreasing order
$a = b^2 - c^2 + d^2$
$\Rightarrow a = b + c + d^2 \Rightarrow d^2 + c = 1$
$\Rightarrow d = 0, c = 1, b = 2, a = 3$.

89. 8204

Sol. $$\sum_{r=1}^{1024} [\log_2 r] = \sum_{r=1}^{2^{10}} [\log_2 r]$$

$$= \sum_{r=2}^{2^2-1} [\log_2 r] + \sum_{r=2^2}^{2^3-1} [\log_2 r] + \sum_{r=2^3}^{2^4-1} [\log_2 r] + \ldots + \sum_{r=2^9}^{2^{10}-1} [\log_2 r] + \log_2 2^{10}.$$

$= 2.1 + 2^2.2 + 2^3.3 + 2^4.4 + \ldots + 2^9.9 + 10$

$$= \sum_{r=1}^{9} 2^r.r + 10 = 8204.$$

90. 2

Sol. $$\int_{-1}^{1} [x[1 + \sin \pi x] + 1]\, dx = \int_{-1}^{0} [x[1 + \sin \pi x] + 1]\, dx + \int_{0}^{1} [x[1 + \sin \pi x] + 1]\, dx$$

Now $-1 < x < 0 \Rightarrow [1 + \sin \pi x] = 0$
$0 < x < 1 \Rightarrow [1 + \sin \pi x] = 1 \Rightarrow [x[1 + \sin \pi x] + 1] = 1$

so $$\int_{-1}^{1} [x[1 + \sin \pi x] + 1]\, dx = 2$$